Tentative Project Outline

Early Sepsis Onset Prediction Using LSTM on ICU Time Series Data

1. Project Goal

The primary objective of this project is to investigate the minimum prediction horizon for reliable sepsis onset prediction using recurrent neural network architectures trained on ICU time-series data.

Core research question of our project:

"How does predictive performance change across different prediction horizons (from 1 hour to 25 hours before sepsis onset), and what is the minimum horizon at which LSTM and GRU models achieve clinically meaningful results?"

This question is clinically significant because earlier detection of sepsis allows faster intervention. Research has shown that each hour of delay in effective antimicrobial therapy reduces survival by approximately 7.6%. By systematically evaluating model performance across different prediction horizons using both standard statistical metrics and the practical Early Detection Advantage (EDA) metric will provide evidence-based guidance on how far in advance reliable predictions can be made.

2. Dataset

In this project I will be using the MIMIC-IV clinical database provided by you. The dataset has been pre-processed to contain 30 hours of ICU measurements per patient admission, structured as follows:

Property	Details
Patient Identifier	stay_id — unique per ICU stay
Time Coverage	30 hours per patient; one measurement per hour
Time Encoding	hours_before column — number of hours before the sepsis decision point (hour 30)
Target Variable	sepsis column — binary (1 = patient develops sepsis within first 30 hours, 0 = no onset)
Missing Values	Present in early hours (NaN); imputed via linear interpolation at patient level; features entirely missing for a patient replaced with the global feature mean
Class Distribution	Imbalanced — sepsis-positive cases are the minority class; requires appropriate handling strategies during training

Feature Set	26 clinically relevant features across four categories: general (5), vital signs (18), diagnosis (2), therapy (1)
Dataset	MIMIC-IV

Sepsis onset prediction and therapy recommendation:

Category	Features
General (5)	Gender, Ethnicity, Age, Height, Weight
Vital Signs (18)	Platelet, Leukocytes, PO2, FiO2, Lactate, Creatinine, Bilirubin, Glasgow Coma Scale, CRP, Diastolic Pressure, Systolic Pressure, Mean Blood Pressure, Respiratory Rate, Temperature, SpO2, Urine Output, Glucose, Heart Rate
Diagnosis (2)	Diabetes, SOFA Score
Therapy (1)	Mechanical Ventilation

3. Methodology

3.1 Data Preprocessing

3.1.1 Missing Value Imputation

Missing values within a patient's time series will be handled using linear interpolation at the patient level. This means that for a given patient, if a measurement is missing at hour (t), its value is estimated by drawing a straight line between the nearest available measurements before and after hour t. This will preserve the trend of the variable rather than simply repeating the last known value.

3.1.2 Feature Aggregation

All features are aggregated into 1-hour time windows following ICU admission to balance temporal resolution with data sparsity. If multiple measurements for a feature occur within the same hour, the most recent measurement will be retained.

3.1.3 Handling Class Imbalance

The dataset is imbalanced with respect to the target variable, with sepsis-positive cases in the minority class. We can evaluate following strategies:

- **Class weights:** Assign a higher penalty to missed sepsis cases (false negatives) during training. Clinically, a missed sepsis case carries far greater cost than a false alarm.
- **Decision threshold adjustment:** Rather than using the default 0.5 classification threshold, tune the threshold using the precision-recall curve to optimize sensitivity for clinical use.

3.2 Data Windowing and Sliding Window Approach

Patient data is structured into overlapping 6-hour input windows following ICU admission. At least 4 to 6 hours of observational data will be required for reliable predictions, and that 5 to 6 hour input windows are particularly effective for early sepsis detection tasks.

The windowing procedure works as follows:

- **Initial window:** Hours 0 to 5 (6 hours of data). The corresponding label indicates whether the patient develops sepsis by hour 30. This yields a prediction horizon of 25 hours.
- **Sliding step:** The window is shifted forward by one hour. Hours 1 to 6 predict sepsis within the next 24 hours. Hours 2 to 7 predict within the next 23 hours, and so on.
- **Final window:** Hours 24 to 29 predict sepsis onset within the next 1 hour, the shortest prediction horizon.

This sliding window approach(we discussed in the previous meeting) will create multiple training samples per patient, each with a different prediction horizon ranging from 25 hours down to 1 hour.

3.3 Baseline Models

Before training recurrent neural networks(RNN), two baseline models will be built using aggregated (non-sequential) features. These serve to: (1) validate that the data pipeline is working correctly, and (2) quantify the added value of sequential modeling. These baselines also serve as the reference point for the Early Detection.

- **Logistic Regression:** Trained on aggregated statistics (mean, min, max) of each feature over the 6-hour window. Provides a linear, interpretable benchmark.
- **Random Forest:** Trained on the same aggregated features. Provides a strong non-linear benchmark without temporal modeling.

3.4 Recurrent Neural Network Models

3.4.1 LSTM : Long Short Term Memory

LSTM is selected as the primary model due to its three-gate mechanism (forget gate, input gate, output gate), which enables strong long-term memory across sequences. This is well-suited for detecting gradual deterioration patterns in patient vital signs.

- **Self-attention layer:** Applied after the input layer to allow the model to assign higher weights to clinically relevant time steps, capturing long-range temporal dependencies.
- **Three LSTM layers:** Each with 16 hidden units. Smaller unit size is appropriate given the 6-hour input window and 26 features.
- **Batch normalization:** Applied after each LSTM layer to stabilize training and accelerate convergence.
- **Dropout:** Rate of 0.2 after each LSTM layer to prevent overfitting.
- **Linear layer:** Reduces dimensionality to 8 units, followed by batch normalization and dropout.
- **Output neuron:** Single linear unit producing a binary prediction (probability of sepsis onset).

3.4.2 GRU — Gated Recurrent Unit

GRU is evaluated as a direct architectural comparison to LSTM. With two gates (reset gate, update gate) and a single combined memory state, GRU has approximately 25% fewer parameters and trains faster while achieving comparable performance on most clinical time series tasks. The architecture mirrors LSTM with the same layer sizes, batch normalization, and dropout configuration; the only structural difference is the use of GRU cells instead of LSTM cells.

3.4.3 Training Configuration

Parameter	Value / Rationale
Loss Function	Binary Cross-Entropy; weighted if class imbalance persists after sampling
Optimizer	Adam — adaptive learning rate, standard for clinical time series
Learning Rate	0.001 (default Adam)
Batch Size	64 patients per update step
Dropout Rate	0.2 after each LSTM/GRU layer
Batch Normalization	Applied after each recurrent layer to stabilize and accelerate training
Early Stopping	Stop if validation loss does not improve for 10 consecutive epochs
Validation Strategy	5-fold cross-validation (80% train / 20% test per fold)
Framework	PyTorch

4. Evaluation

4.1 Standard Metrics

Metric	What It Measures	Why It Matters Here
AUC (AUROC)	Model's ability to separate sepsis from non-sepsis across all thresholds	Threshold-independent and robust to class imbalance; exceeding 0.80 is the clinically relevant threshold cited in the literature
F1-Score	Harmonic mean of precision and recall	Captures both false alarms and missed detections, both of which carry significant clinical cost in ICU settings
Sensitivity	Fraction of sepsis cases correctly detected	Critical so for every missed sepsis case risks a 7.6% reduction in hourly survival
Specificity	Fraction of non-sepsis cases correctly cleared	Too many false alarms cause alarm fatigue in clinical staff, reducing trust in the system
AUPRC	Area under Precision-Recall curve	More informative than AUC when positive class is rare the preferred metric for imbalanced datasets

4.2 Practical Evaluation — Early Detection Advantage (EDA)

Beyond standard metrics, in this project we can adopt an EDA-based evaluation approach inspired by your 2025 paper. In that paper, EDA was used to measure how many hours earlier the federated model detected sepsis compared to local baselines. In this project, EDA is adapted to a single-institution setting: it will measure how many hours earlier the LSTM or GRU detects sepsis compared to the classical baseline models (Logistic Regression and Random Forest).

EDA directly answers the most clinically relevant question of this project:

- **Does using a sequential model (LSTM or GRU) allow doctors to be warned about sepsis onset earlier than using a simple static model?**
- **By how many hours, on average, can the warning be issued sooner?**
- **Which architecture — LSTM or GRU — provides the greater early detection advantage?**

4.5 Summary of Evaluation Framework

Perspective	Method	Key Question Answered
Statistical	AUC, F1, Sensitivity, Specificity, AUPRC	How accurate is the model overall?
Practical / Clinical	Early Detection Advantage (EDA)	How many hours earlier does it detect sepsis vs baseline?

5. Technology Stack

Category	Tool / Library	Purpose
Language	Python	Primary programming language for all experiments
Deep Learning	PyTorch	LSTM and GRU model implementation, attention layer, training loop
Classical ML	scikit-learn	Logistic Regression and Random Forest baseline models
Data Handling	pandas, numpy	Data loading, linear interpolation, feature engineering, windowing
Imbalance	imbalanced-learn	SMOTE implementation for class rebalancing if needed
Evaluation	scikit-learn metrics	AUC, F1, AUPRC, sensitivity, specificity computation
Visualization	matplotlib	Performance-over-time curves, EDA bar charts, temporal analysis plots
Version Control	Git / GitHub	Code management, reproducibility, submission
Environment	conda / venv	Dependency and environment management

6. Expected Outcomes

- **Temporal performance profile:** A curve showing how LSTM and GRU performance (AUC, F1) changes across prediction horizons from 1 to 25 hours, identifying where performance degrades and where it is stable.
- **Minimum reliable horizon:** An evidence-based answer to how far in advance LSTM and GRU can reliably predict sepsis onset with clinically meaningful accuracy.
- **Early Detection Advantage:** A quantified measure of how many hours earlier the recurrent models detect sepsis compared to classical baselines the most practically meaningful output.
- **LSTM vs GRU comparison:** A direct comparison of both architectures across all prediction horizons in terms of both statistical and practical performance.
- **Baseline comparison:** Quantification of the value added by sequential modeling over classical static approaches, showing where and by how much RNNs outperform simpler models.
- **Optional extension:** TimeSHAP analysis explaining which features and time steps drive predictions at the most clinically relevant horizons.

7. Proposed Timeline (Tentative)

Phase	Duration	Tasks
Phase 1	Weeks 1–3	Data loading, linear interpolation, feature normalization, sliding window construction, train/test split
Phase 2	Weeks 4–5	Baseline models: Logistic Regression and Random Forest on aggregated features; validate pipeline correctness
Phase 3	Weeks 6–8	LSTM and GRU implementation with attention layer, batch normalization, and dropout; hyperparameter tuning via 5-fold CV
Phase 4	Weeks 9–10	Evaluation: AUC, F1, EDA, temporal performance analysis, baseline improvement curves; all reported mean ± std across folds
Phase 5	Weeks 11–12	(Tentative)TimeSHAP feature importance analysis on best-performing model at optimal prediction horizon
Phase 6	Weeks 13–14	Final report writing, visualization, documentation, GitHub repository preparation